

LUIS JAVIER LOZOYA

Senior Application Security Engineer | AI/LLM Security

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PROFESSIONAL SUMMARY

Software engineer with 5+ years of experience building cloud-native applications, with hands-on experience integrating application security throughout development, including secure code review, authentication and authorization, OWASP Top 10, threat modeling, and AWS security. Author of llm-audit, an open-source Semgrep rule pack mapped to the OWASP Top 10 for LLM Applications, with hands-on experience in LLM application security and security automation. GIAC GFACT, GSEC, and GCIH certified through the SANS Cyber Academy and HarborHack 2025 conference speaker. Bilingual EN/ES.

COMPETENCIES & SKILLS

AI/LLM Application Security: OWASP Top 10 for LLM Applications (LLM01, LLM02, LLM03, LLM10 hands-on via llm-audit); secure prompt design; familiar with RAG and vector database security patterns, tool-calling abuse, and Model Context Protocol (MCP) tooling

Application Security: OWASP Top 10, threat modeling, SAST/DAST, secure code review (TypeScript, JavaScript, Python), authentication and authorization design, input validation, security headers, Secure SDLC, SOC 2 Type 1 control evidence collection

Cloud Security: AWS IAM (least privilege), AWS Cognito, KMS, Secrets management, VPC isolation, S3 bucket policies, CloudFront, Lambda hardening

Assessment Tools: Burp Suite, Postman, Semgrep, sqlmap, DirBuster, jwt_tool, Wireshark, tcpdump, nfdump/nfpcapd

Familiar (SANS labs): Metasploit, Nmap, OpenSSL

LLM APIs & Integrations: OpenAI SDK, Anthropic Claude, prompt engineering

Languages & Frameworks: TypeScript, JavaScript, Python, Node.js, React, Next.js, Angular, Svelte, Express, Tailwind

Cloud & Data Platforms: AWS (Lambda, Amplify, Cognito, S3, CloudFront, IAM); Azure (Functions, Pipelines); GitHub Actions; MongoDB; Cosmos DB; Supabase (Postgres)

EDUCATION & CERTIFICATIONS

SANS Cyber Academy Scholarship, intensive cybersecurity training program **Fall 2025 to Present**

Earned: GIAC GFACT (Jan 2026), GIAC GSEC (Apr 2026), GIAC GCIH (Aug 2026). Coursework: SEC275, SEC401, SEC504.

In progress: PortSwigger BSCP (target Q4 2026); AWS SCS-C03 (target Q1 2027);

Cybersecurity Path, System Administration, Purdue University Northwest **March 2022 to May 2023**

Full Stack Developer, JRS Coding School **November 2019 to October 2020**

B.S. Architectural Engineering, IE University **August 2006 to March 2012**

CYBER PROJECTS AND ENGAGEMENTS

llm-audit

- Authored an open-source Semgrep rule pack mapped to the OWASP Top 10 for LLM Applications to identify security weaknesses in TypeScript and JavaScript LLM applications.
- Integrated the rule set into an npm CLI, Husky pre-commit hook, and GitHub Actions workflow to automate security testing within the development lifecycle.
- Identified and remediated an LLM10 Improper Output Handling vulnerability in the recruiter-fit endpoint on luislozoya.com during application testing.

HackOps 2024, First Place: HarthHaven (*senior care discovery application*)

- Built a Svelte and Figma application with a team of four to help family caregivers select appropriate senior communities.
- Contributed data set construction for the demo application.

AWS VPC Flow Log Analysis (*SANS Cyber Academy lab*)

- Triaged 173K flow records from 579 compressed log files using nfdump and nfdump.
- Identified 265 MB of exfiltration over non-standard port 8889.
- Correlated attacker activity across a 6.5-hour window.

SANS Core NetWars Tournament 11 & 12 (*SANS flagship CTF*)

- Completed two consecutive SANS Core NetWars tournaments covering vulnerability discovery, exploitation, defense, and forensics.
- Multi-level CTF format testing offensive and defensive security skills across enterprise scenarios.

EXPERIENCE

Security Focused Software Engineer, g/d/n/a

March 2024 to Present

- Conducted security review and dependency analysis on hybrid AWS and MongoDB architectures, surfacing exposure paths before deployment.
- Designed and implemented secure authentication and authorization flows using AWS Cognito, enforcing session management, input validation, and security headers across production applications.
- Architected least-privilege IAM policies across 15+ AWS resources (Lambda, S3, API Gateway, RDS), reducing blast radius in the event of credential compromise.
- Delivered React applications on AWS Lambda and Amplify with production-grade security controls (encrypted environment variables, IAM-scoped function roles, signed CloudFront distributions).
- Supported SOC 2 Type 1 readiness by collecting control evidence across AWS infrastructure, coordinating with internal engineering and operations teams, reviewing infrastructure configurations, and contributing to security policy documentation.

Software Engineer, Querri

August 2023 to March 2024

- Implemented FusionAuth-based authentication with secure session handling, role-based access controls, and audit logging for a Svelte-based product application.
- Hardened front-end and embedded code paths against common web vulnerabilities (XSS, CSRF, insecure direct object references) during feature development.
- Developed and maintained custom HubSpot CMS modules and templates supporting the product marketing site alongside core engineering.

Software Engineer II, Interloop Technologies, Inc.

July 2021 to June 2023

- Led Angular, NestJS, and MongoDB projects with attention to authentication, authorization, and input validation in API design.
- Built and operated Azure Functions for client automation, integrating with downstream Azure services and CRM APIs.
- Mentored junior engineers through security-aware code reviews and pair programming.

Business Developer, Project Engineer, Field PM, Coastal Millwork and Supply

September 2015 to June 2021

- Delivered commercial millwork projects across multiple roles, including the Google Charleston Office.
- Managed project timelines, vendor coordination, and on-site installations for high-end architectural millwork.
- Transitioned into software engineering via JRS Coding School in 2019.